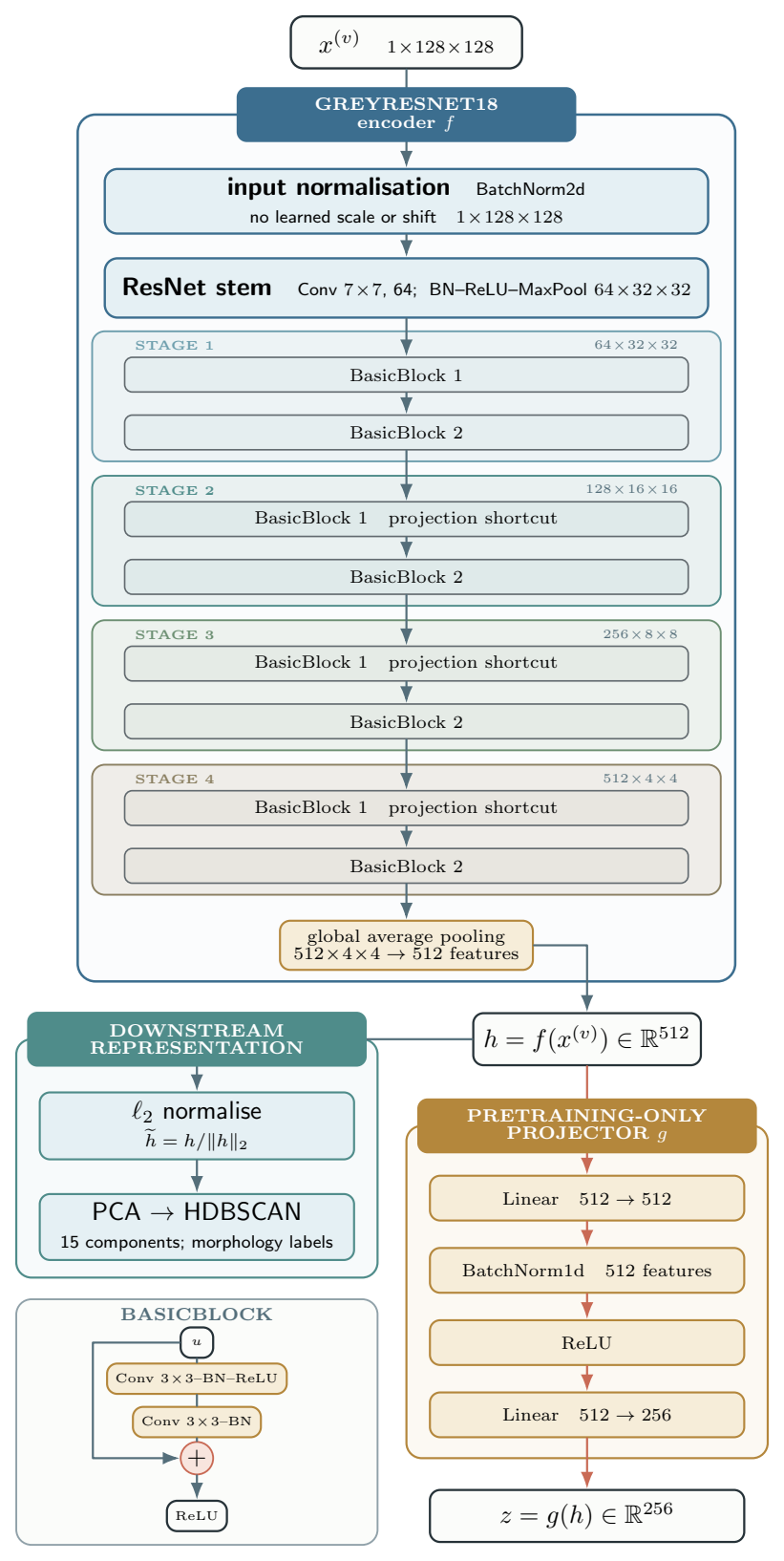

$x^{(v)}$ $1\times128\times128$
GREYRESNET18
encoder f
input normalisation BatchNorm2d
no learned scale or shift $1\times128\times128$
ResNet stem Conv 7×7, 64; BN–ReLU–MaxPool $64\times32\times32$
STAGE 1
$64\times32\times32$
BasicBlock 1
BasicBlock 2
STAGE 2
$128\times16\times16$
BasicBlock 1 projection shortcut
BasicBlock 2
STAGE 3
$256\times8\times8$
BasicBlock 1 projection shortcut
BasicBlock 2
STAGE 4
$512\times4\times4$
BasicBlock 1 projection shortcut
BasicBlock 2
global average pooling
$512\times4\times4 \to 512$ features
$h = f(x^{(v)}) \in \mathbb{R}^{512}$
DOWNSTREAM REPRESENTATION
ℓ_2 normalise
$\tilde{h} = h/\|h\|_2$
PCA $\to$ HDBSCAN
15 components; morphology labels
PRETRAINING-ONLY PROJECTOR g
Linear $512 \to 512$
BatchNorm1d 512 features
ReLU
Linear $512 \to 256$
$z = g(h) \in \mathbb{R}^{256}$
BASICBLOCK
u
Conv 3×3–BN–ReLU
Conv 3×3–BN
ReLU